Is opacity enough? Angular relaxation spectra of the QGP

Olga Soloveva (ITP & GSI)

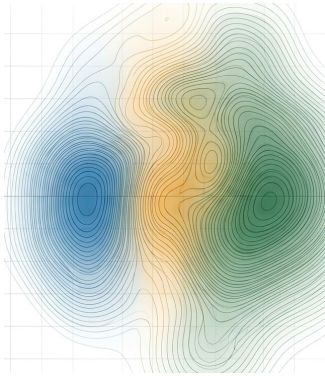
Based on

arxiv: 2609.05703

22 October 2026

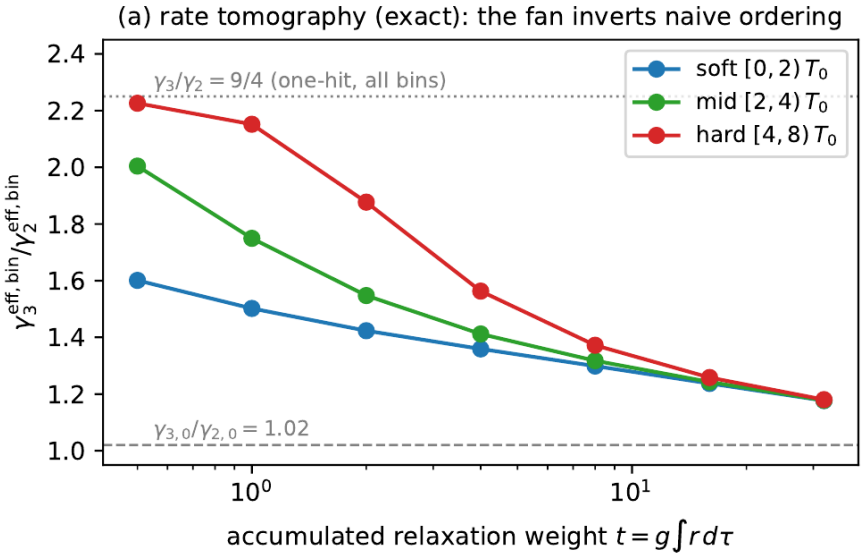
Transport and Magnetohydrodynamics meeting

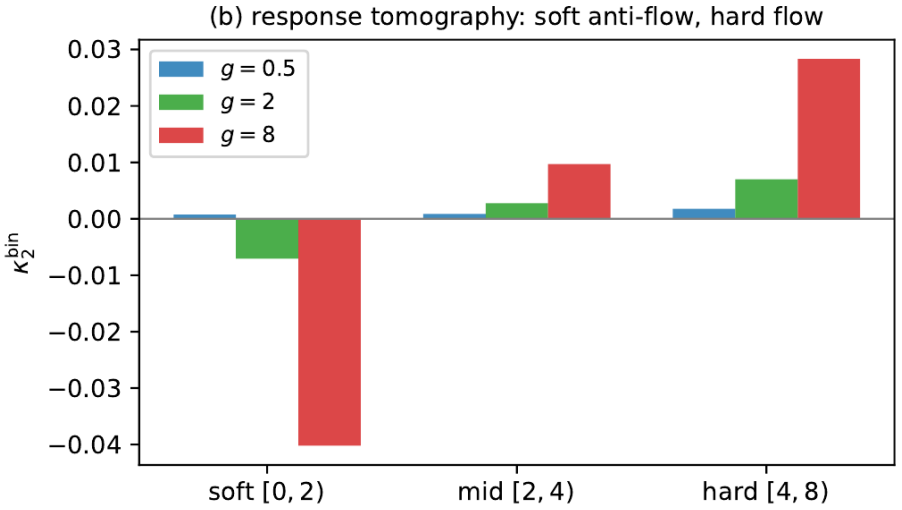
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CRC-TR 211 Strong-interaction matter under extreme conditions

Small systems flow, and kinetic theory must describe them

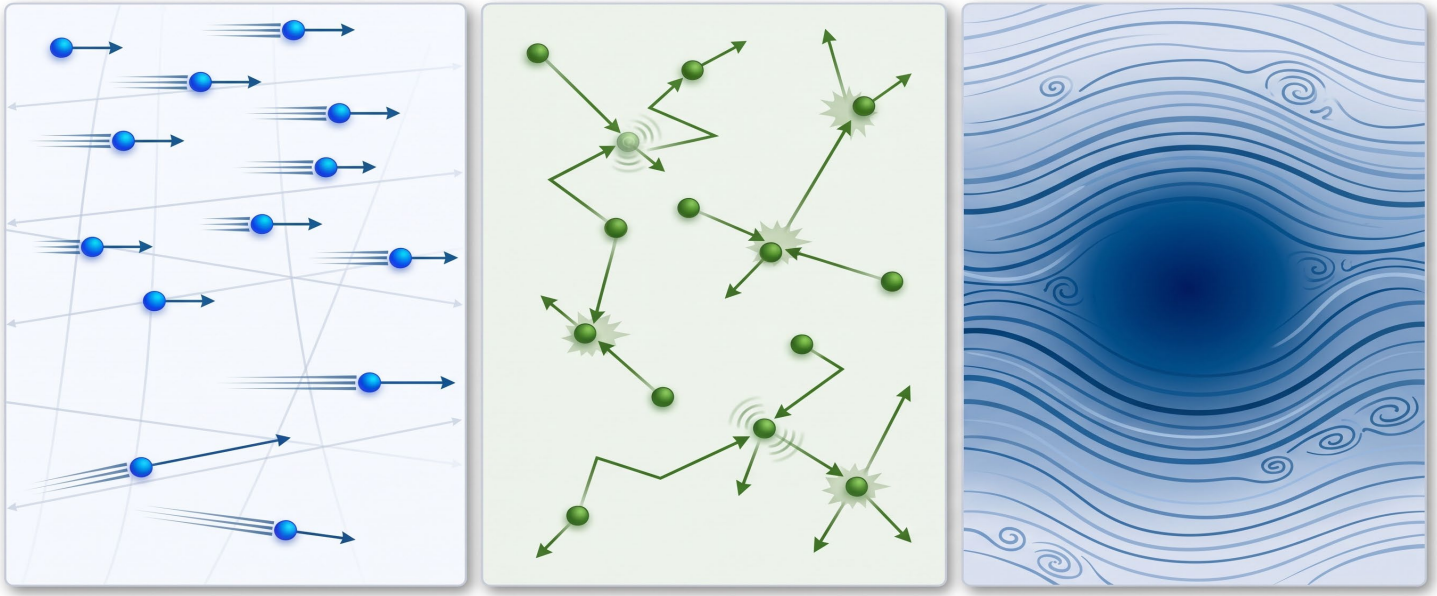




- Horizontal axis: accumulated opacity t = g ∫ r dτ
- All momentum bins start at the one-hit value 9/4 and approach 50/49 at large opacity
- Left: effective ratio γ3/γ2 per momentum bin. Right: the flow carried by each bin

What does the flow response carry beyond η/s?

Small systems flow, and kinetic theory must describe them



Low g: free streaming | g ~ 1: kinetic | High g: fluid

What is opacity?

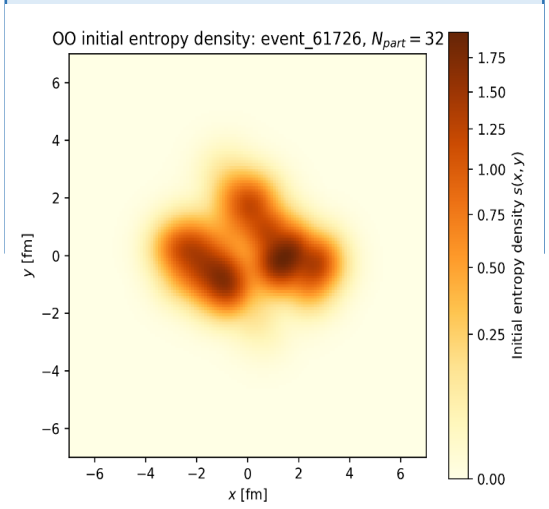
Opacity g is a dimensionless inverse Knudsen number.

It sets how fast the thermal second harmonic relaxes, in units of the system size.

Small systems flow, and kinetic theory must describe them

Observed

Collective flow in p+p, p+Pb and O+O collisions



Kinetic picture

At finite opacity, non-hydrodynamic modes remain important

Open question

Models matched to the same η/s predict different v_n at intermediate opacity

What does the flow response carry beyond η/s?

Small systems flow, and kinetic theory must describe them

Dynamical regimes (C. Werthmann et al.)

Large systems: a hydrodynamic QGP.
Light ions: few scatterings, possibly not fully equilibrated.
Kinetic theory covers free streaming to hydrodynamics.
arXiv:1905.05139 (Kurkela, Wiedemann, Wu)

Early times (S. Schlichting et al.)

Pre-equilibrium dynamics sets the initial state of hydrodynamics.
Collisions equilibrate on times $\ll$ 1 fm/c, before hydro applies at ~1 fm/c.

arXiv:1805.01604, 1805.00961 (Kurkela et al.)

System size (C. Frascà et al.)

Hydrodynamization depends on system size, studied in kinetic theory.
Light-ion v_n data for O+O and Ne+Ne are now available (ALICE, CMS).

Centrality (A. Kirchneret al.)

Centrality estimators differ by 2 to 4 percentage points.
The ε_n and v_n bins must use one consistent definition.

Light ions are where the relaxation spectrum decides which kinetic description applies.

Why it matters?

Bayesian fit

Global fits constrain transport coefficients mainly through η/s.

Mode-resolved parameters, such as the separation $A + n^2B$, could enter future Bayesian analyses as extra constraints beyond viscosity.

Cold atoms

Ultracold Fermi gases and 2 D electron fluids show the same separation between hydrodynamic transport and microscopic relaxation.

Cold-atom experiments already measure relaxation toward attractors (Fujii and Enss, PRL 2024).

arXiv:2405.03635
arXiv:2408.02738

Early stages

Before hydrodynamics applies, the response is set by how the higher angular modes relax.

Small systems keep this sensitivity, which is where the early stage matters most.
arXiv:1805.01604 (Kurkela et al.)
Giacalone et al., PRL 123, 262301 (2019

Common question: which microscopic relaxation rates control the flow response when the system is not in local equilibrium?

What is opacity?

$$Kn = \lambda / L$$

$$\text{opacity } g \sim 1 / Kn = L / \lambda$$

λ: mean free path between collisions
L: size of the system

Opacity is the inverse Knudsen number: how many relaxation times fit into the system's lifetime.

Kn ≫ 1 (g small)

Rare collisions. Particles stream freely, so the flow is weak and kinetic theory is needed.

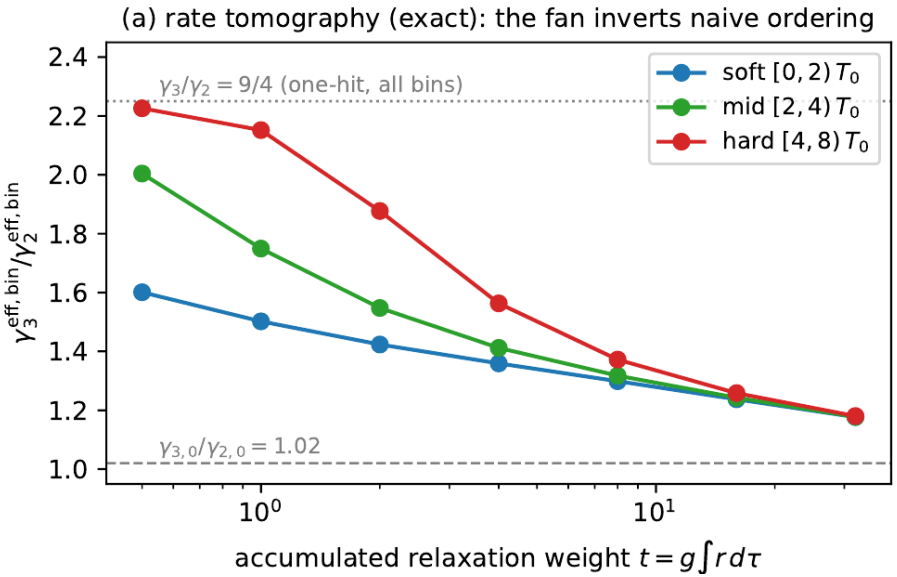
Kn ~ 1 (g ~ 1 to 10)

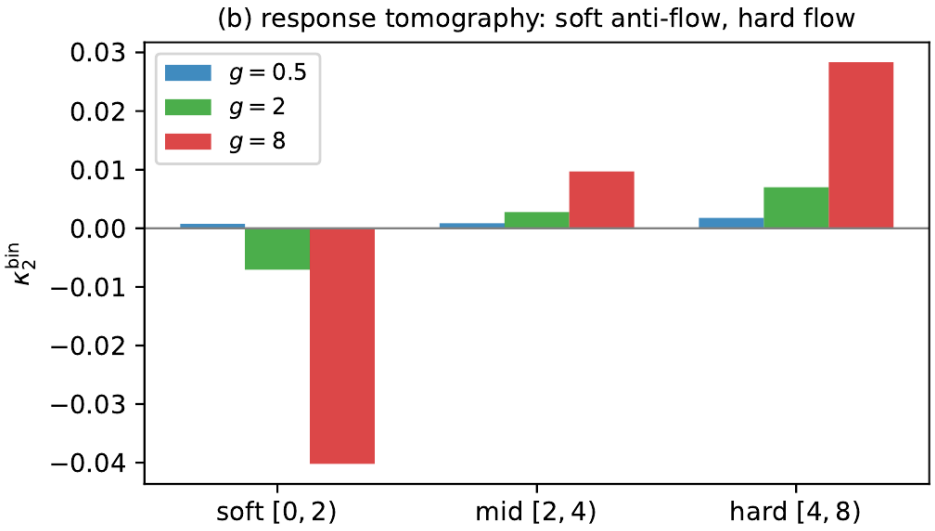
Intermediate opacity. This is the regime studied in this talk, where the microscopic details matter.

Kn ≪ 1 (g large)

Frequent collisions. Local equilibrium holds and hydrodynamics with viscosity works.

Small systems flow, and kinetic theory must describe them





- Horizontal axis: accumulated opacity t = g ∫ r dτ
- All momentum bins start at the one-hit value 9/4 and approach 50/49 at large opacity
- Left: effective ratio γ3/γ2 per momentum bin. Right: the flow carried by each bin

What is γ? The relaxation rate of an angular harmonic

$$C[\Phi] = -\Sigma \ell\, \gamma\ell\, (\Phi\ell - \Phi\ell\text{^eq})\, e\text{^}\{i\ell\varphi\}$$

- γℓ: how fast the ℓ-th angular harmonic of the particle distribution relaxes to local equilibrium
- Units: inverse time. Its inverse 1/γℓ is the relaxation time of that harmonic
- Larger γℓ: collisions damp that harmonic faster
- Each harmonic ℓ has its own rate γℓ, which is the central object of this talk

Conservation fixes two rates to zero

$$\gamma_0 = \gamma_1 = \gamma_{-1} = 0$$

Why zero?

$\gamma_0 = 0$: energy is conserved.

$\gamma \pm 1 = 0$: momentum is conserved.

A naive angular-diffusion term has $\gamma_1 \neq 0$ and breaks momentum conservation.

Scalar rates

For a scalar kernel, $\gamma_\ell = \hat{g}_\ell \, W(x,\tau)$.

$W = e^{\alpha}$ sets the medium dependence: $\alpha = 1$ for a rate $\propto$ density, $\alpha = 1/3$ for a conformal rate $\propto T$.

The ratio of two rates, $\gamma 3/\gamma 2$, is independent of W.

Only the ratios of the $\hat{g}_\ell$ are fixed by the microscopic collision kernel.

Conservation fixes two rates to zero

$$\Phi n(\infty) = \int d\tau\, Sn(\tau) \exp[- \int \tau \hat{} \infty\, \gamma n\, d\tau']$$

Low opacity

Few collisions: the flow response κn grows with γn. Scattering is what creates the anisotropy.

High opacity

Many collisions: the response is damped by exp(−γn). The flow is suppressed, a kinetic analogue of the Gurzhi crossover.

Hydrodynamic limit: $\eta \propto (e + P)/\gamma_2$ and $\tau\pi = 1/\gamma_2$. Only the $\ell = 2$ rate enters.

Viscosity only sees the $\ell = 2$ angular sector

- The drift term couples harmonic m only to m ± 1
- First-order Chapman–Enskog closes on the $\ell = \pm 2$ channel
- $\eta \propto (e + P)/\gamma_2$ and $\tau\pi = 1/\gamma_2$
- Rates $\gamma_{\ell \geq 3}$ first enter at third order in gradients

$$\gamma_2 \, \delta\Phi\pm 2 = -\tfrac{1}{2} \, (\partial x \mp i\partial y) \, \Phi\pm 1$$

Any two kernels with the same γ_2 share η and ττ, while their higher rates can still differ.

Three questions for this talk

1. Does matching η/s fix the flow response at finite opacity?

2. Which relaxation rates do κ_2, κ_3 and their ratio measure?

3. How much microscopic information survives when the system does not reach local equilibrium?

Model: mode-resolved relaxation-time approximation

$$C[\Phi] = -\Sigma \ell\, \gamma\ell\, (\Phi\ell - \Phi\ell eq)\, e^{\{i\ell\varphi\}}$$

$$\gamma_0 = \gamma\pm_1 = 0 \quad \text{(conservation)}$$

- Scalar case: γℓ = ĝℓ W(x,τ), with W = e^α
- α = 1: rate ∝ density; α = 1/3: conformal rate ∝ T

Kernel	Rate γℓ
Flat RTA	1/τR, all ℓ ≥ 2
Angular diffusion	D ℓ² (breaks momentum)
Momentum-conserving	D (ℓ² − 1)
Mixed spectrum	a + b ℓ²

At leading opacity, flow ratios measure rate ratios

$$(\kappa n/\kappa m)\text{spec} \div (\kappa n/\kappa m)\text{flat} = \gamma n / \gamma m$$

- Exact at leading order in opacity
- Independent of the initial eccentricity profile, of α, and of the equilibrium subtraction
- Only the spectrum enters, through the prefactor $\hat{g}n$
- Double ratio against flat RTA: D_nm

Dilute anchors: the double ratio fingerprints the spectrum

Spectrum (2D)	γ3/γ2	γ4/γ2	D32	D42
Flat RTA	1	1	1	1
Angular diffusion, D ℓ^2	**9/4**	**4**	**9/4**	**4**
Momentum-conserving, D(ℓ^2−1)	**8/3**	**5**	**8/3**	**5**
Mixed, a + b ℓ^2	(a+9b)/(a+4b)	—	1 to 9/4	—

Finite opacity: the signal decays slowly

9/4 → 1.92

D_{32} of angular diffusion, from one hit to $g = 8$

$g \gtrsim 130$

needed for $|D_{32} - 1| < 0.1$ (code opacity parameter)

3 decades

of opacity over which higher-mode sensitivity persists, from p+p to Pb+Pb

The rate that generates v_n at low opacity also damps it after repeated scattering.

Beyond scalar rates: QCD Fokker–Planck kernel

Leading-logarithmic QCD transport

The leading-log QCD kernel is not a scalar

$$C_FP[f] = \kappa \nabla p \cdot (T \nabla p\, f + p\, f)$$

$$\Lambda \ell = -\kappa \left[\partial p^2 + (1/p + 1/T)\partial p + 1/(pT) - \ell^2/p^2\right]$$

$$\gamma \ell(p) = \kappa\, \ell^2 \,/\, p^2$$

- Each harmonic block is a differential operator in momentum
- The local angular rate is strongest for soft partons
- The angular index and momentum modulus are entangled
- No choice of $\hat{g}\ell$ W reproduces this operator
- $\ell = 0, \pm 1$ stay frozen, so conservation holds

A closed-form spectrum: a hydrogen-like problem

$$\gamma_{\ell,k} = \kappa/(4T^2) \cdot [1 - 1/(2\ell + 2k + 1)^2]$$

$$\text{continuum edge: } \gamma \geq \kappa/(4T^2)$$

- Liouville map to a radial Coulomb Hamiltonian
- Soft momenta form the continuum, which carries the ℓ^2 hierarchy
- Hard momenta form bound states, nearly insensitive to ℓ
- Degeneracy: $\gamma_{\ell,k} = \gamma_{\ell+1,k-1}$
- Same Fokker–Planck–Coulomb correspondence as Gavassino's quasinormal-mode analysis

50/49

$\gamma_{3,0} / \gamma_{2,0} \approx 1.02$, compared with 9/4 for naive angular diffusion

Three observables probe three projections of one operator

1

dilute response: the thermal one-hit matrix element

1/4

viscosity: an inverse spectral moment, with η_FP / η_flat = 4

0.12

late-time decay: the lowest surviving mode γ2,0

A flat RTA matched to η has four times the Fokker–Planck dilute response and eight times its asymptotic decay rate.

Momentum-resolved tomography: repeated scattering reorders modes

- The effective ratio γ3eff/γ2eff starts at 9/4 in every momentum bin
- Soft bins reach the bound-state regime first and approach 50/49
- Hard bins keep the continuum hierarchy longest
- The flow-carrying weight shifts toward larger momenta
- In a frozen bath the soft contribution can change sign, which no scalar spectrum can do

The bath temperature changes the crossings, not the separation

Frozen bath, T = T0

κ2 and κ3 cross zero at intermediate opacity (g3* between 2 and 3 for α = 1).

The crossings diagnose the rate model rather than a physical prediction.

Local conformal T(x,τ)

The crossings disappear; κ3 stays positive through g = 4.

D32 plateau near 3/2, distinct from flat RTA (1) and the scalar ℓ^2 spectrum (~ 2.1 to 2.2).

The harmonic separation is robust against the choice of bath temperature.

SolKin: a deterministic kinetic solver

Boltzmann equation with angular decomposition

$$\Phi(x,\phi,\tau) = \Sigma_\ell \, \Phi_\ell(x,\tau) \, e^{i\ell\phi}$$

- ❑ Exact spectral advection in space at fixed momentum angle
- ❑ Exact per-harmonic exponential relaxation toward the Landau-matched equilibrium
- ❑ Energy and momentum conserved to 10^{-14}
- ❑ Results stable under changes of p_min, p_max, boundary condition and radial grid
- ❑ Production grid: N_x = 80, N_φ = 32, N_p = 20, p_max = 12 T_0
- ❑ Publicly available on Zenodo

Current setup:
massless partons in the transverse plane
3 types of Collision kernels:

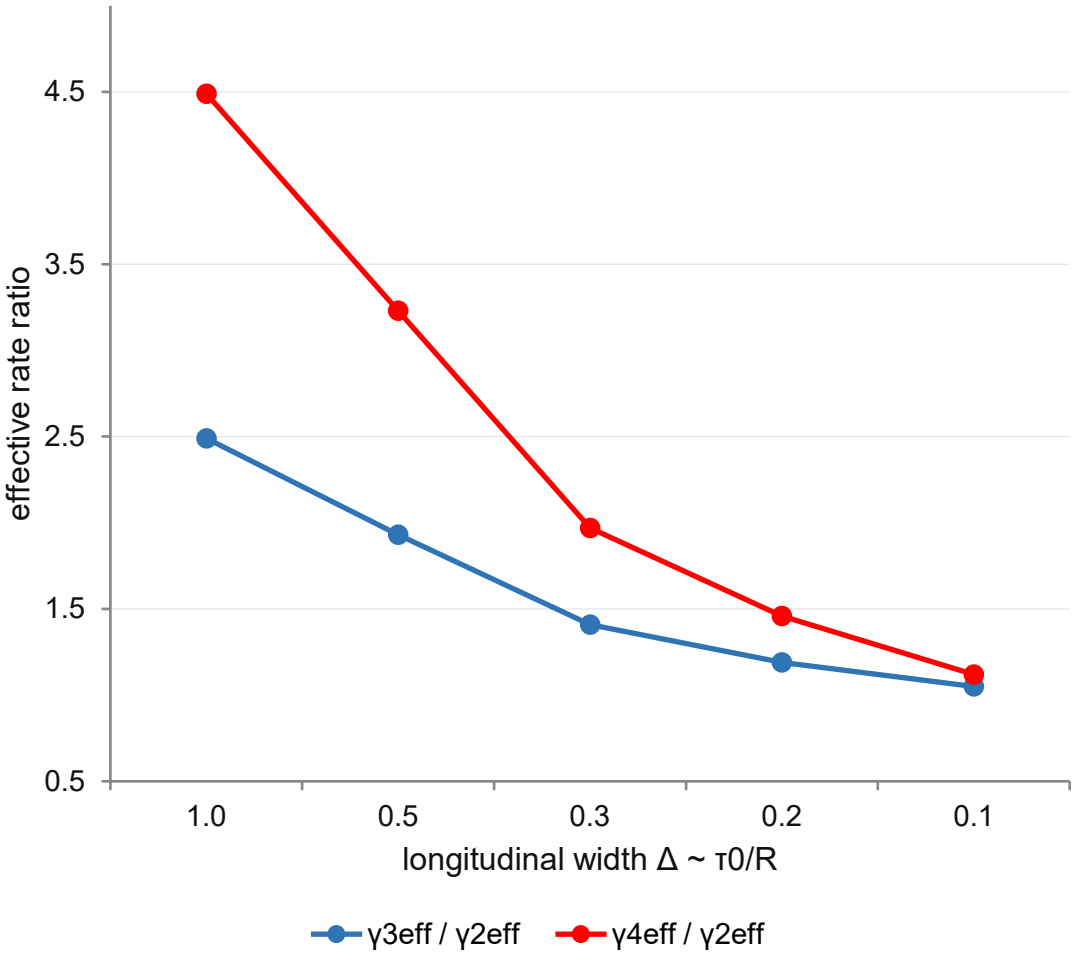
- flat RTA (conventional RTA)
- MRTA/angular spectra
- momentum-dependent Fokker–Planck kernels.

Key distinction
η/s probes the ℓ = 2 sector, while finite-opacity flow can retain information from ℓ ≥ 3.

Observable

$$\kappa_n = v_n / \varepsilon_n \quad \text{and} \quad D_{nm}$$

Small systems - keep the spectrum; large systems - wash it out



- Longitudinal expansion mixes the angular and azimuthal labels
- For small **Δ: γn^eff/γ2^eff ≃ 1 + (n² − 4)Δ²**
- Small systems with **τ0/R** = O(1) keep full sensitivity
- Large systems lose it hydrodynamically and through the longitudinal squeeze

SMASH

- SMASH (Simulating Many Accelerated Strongly-interacting Hadrons) is a hadronic transport model
- It propagates hadrons and their binary collisions and decays with no hydrodynamic stage
- In the 'pure SMASH' description, the whole evolution, including the dense early stage, is hadronic
- Its response is therefore set by the hadronic scattering rates

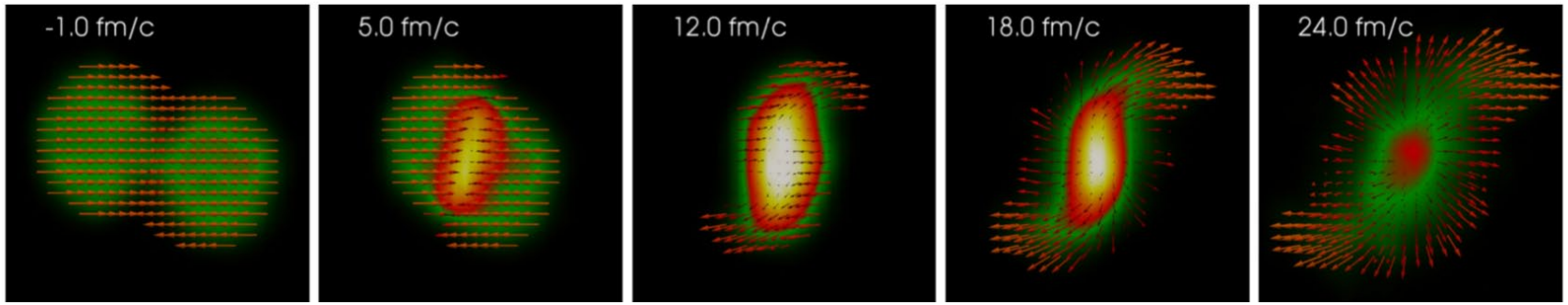
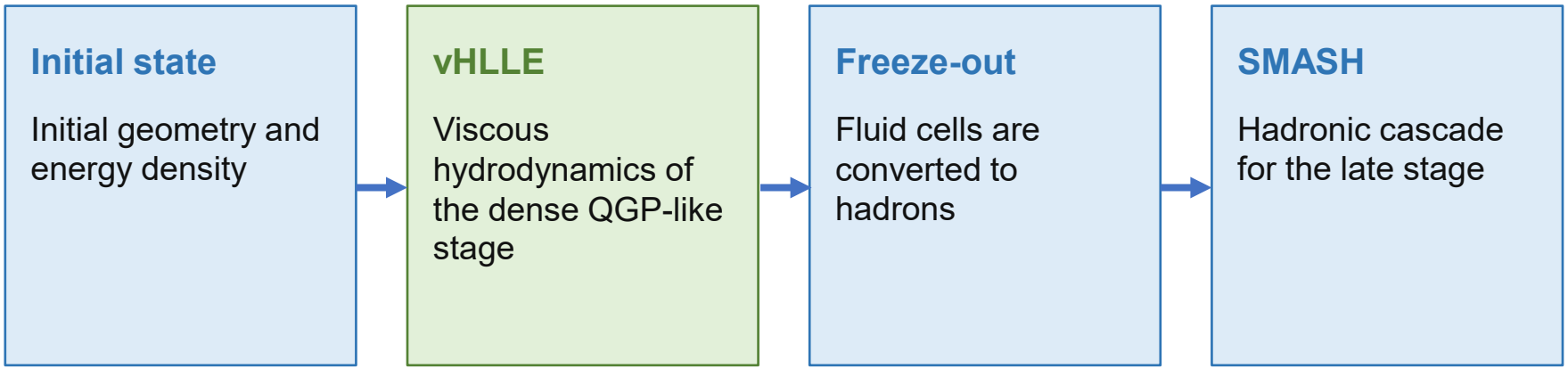


Illustration is taken from talk H.Elfner

Role in this study

The microscopic reference: its κ_2, κ_3 and R_{32} are the data that the SolKin kernels are compared with.

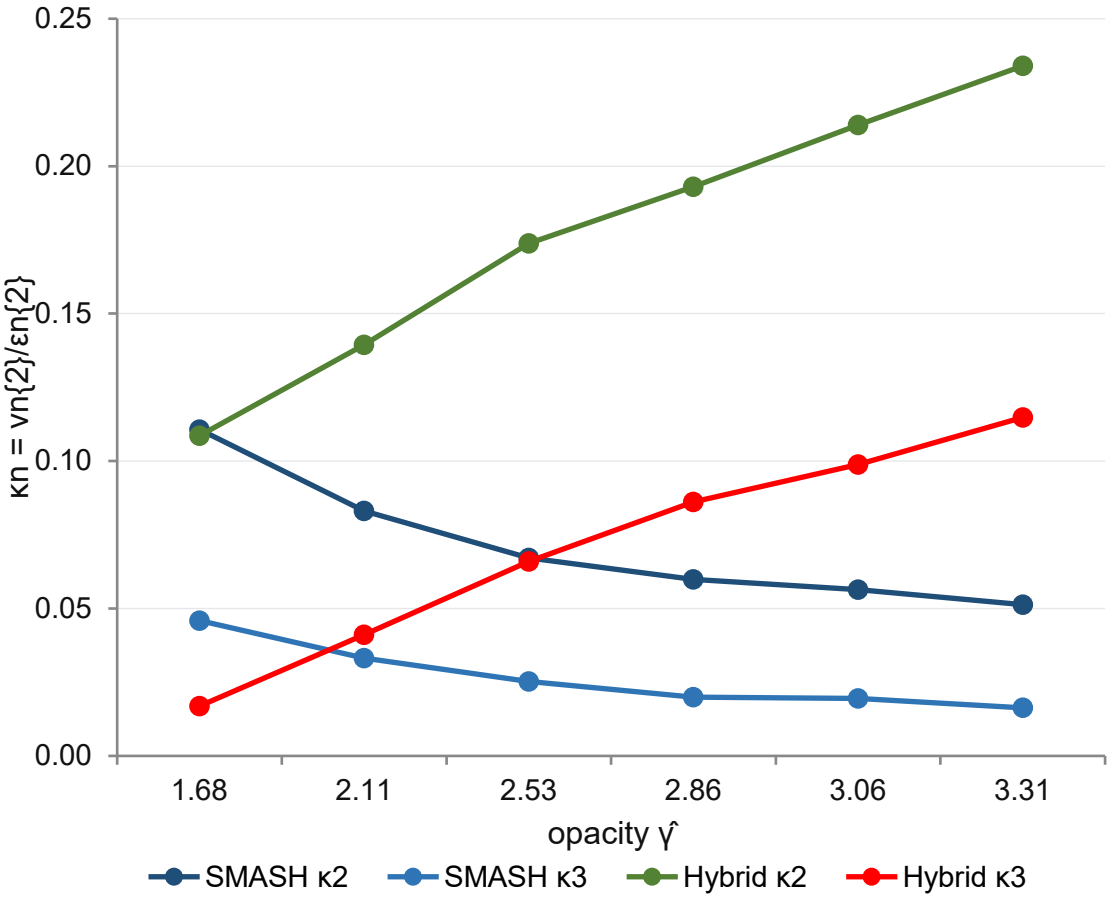
SMASH–vHLLE: a hybrid of hydrodynamics and transport



- ❑ Hydrodynamics describes the dense stage, where a fluid description applies
- ❑ The hadronic cascade describes the dilute late stage, where a fluid description fails
- ❑ The response is therefore sensitive to where the switch happens

References: arXiv:2112.08724 (SMASH–vHLLE hybrid); arXiv:2503.10181 (Bayesian analysis of the (3+1)D hybrid)

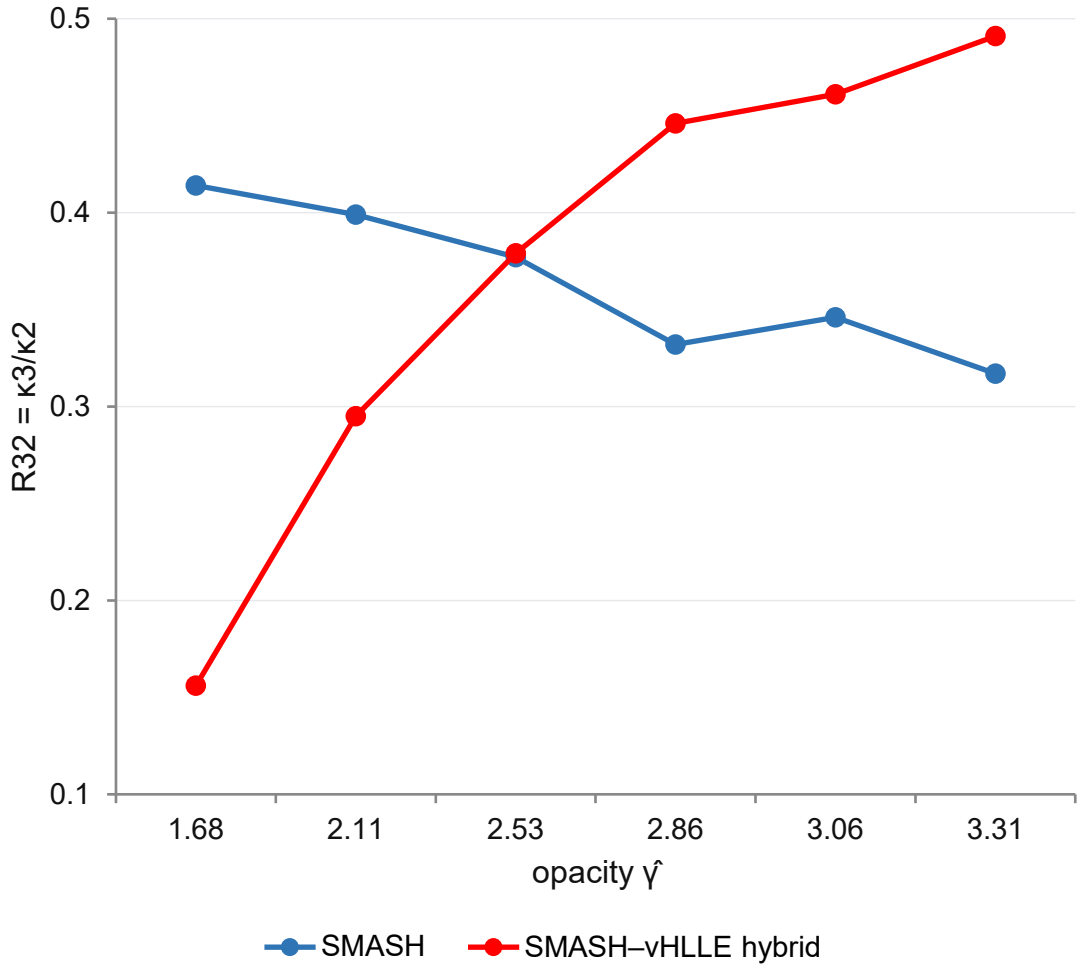
SMASH O–O: response coefficients at matched opacity



- O–O at √sNN = 5.36 TeV, Woods–Saxon
- Pure SMASH: κ2 and κ3 decrease with opacity
- SMASH–vHLLE: response grows strongly with opacity
- Same opacity, different harmonic response

Data source: arXiv:2509.05613 and its public ancillary files

The ratio R32 = κ3/κ2 separates the two models



- SMASH falls from 0.41 to 0.32; the hybrid rises from 0.16 to 0.49
- The curves cross near $\hat{\gamma} \approx 2.5$
- The ratio reduces sensitivity to overall normalization
- One point alone cannot identify the dynamics

SolKin comparison against SMASH

1. Linear response for n = 2 (β2 = 0.04) and n = 3 (β3 = 0.01)

2. Scan g for flat RTA, angular-diffusion MRTA and the momentum-dependent Fokker–Planck kernel

3. Fix g = Cg $\hat{\gamma}$ from one dilute n = 2 point, then keep Cg for n = 3

4. Compare κ2(g), κ3(g), R32(g) and D32 against flat RTA

Which SolKin spectrum best reproduces the microscopic SMASH response?

Applicability to small systems

- SolKin needs only the initial geometry and the opacity as inputs
- The same analysis therefore applies to SMASH results for O–O and p–A collisions
- Opacity ranges overlap for O–O (Woods–Saxon), clustered O–O and clustered Ne–Ne (about 1.7 to 3.5), enabling a geometry test
- Geometry-dependent final-state effects can be quantified kernel by kernel

What is still preliminary

- ε_n{2} comes from event-by-event eccentricities by centrality percentile, paired with published integrated v_n{2}
- An event-level mapping to the final-state multiplicity estimator is not yet done
- Complete integrated v2 and v3 for the clustered systems are still to be assembled
- Current numbers are inputs for designing the comparison, not final precision results

Outlook: four directions

Mode-resolved hydrodynamization

Extend the inverse-Reynolds analysis to $\ell = 3, 4$ and test whether η/s matching suffices.

Final-state effects in Ne–Ne / O–O

$\kappa 2 \propto \hat{\gamma}^{1.1}$ and $\kappa 3 \propto \hat{\gamma}^{2.8}$: a 5% opacity difference could shift ratios by about 6% and 15%.

Real-data test: STAR d+Au vs O+O

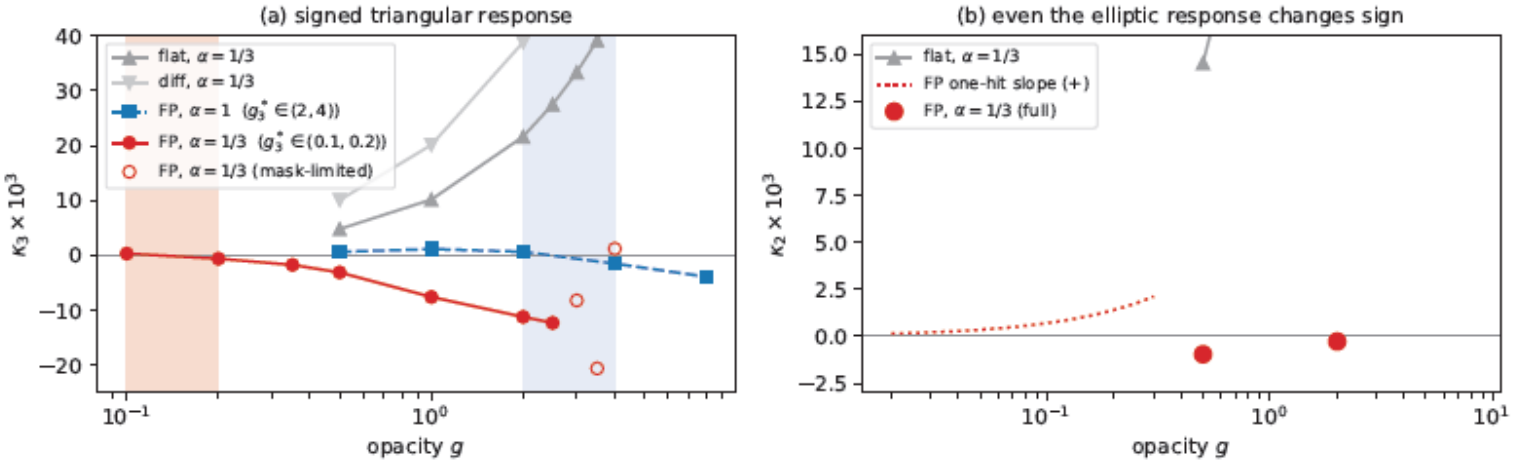
Can one $\kappa n(g(N_{ch}))$ relation with one kernel describe two geometries at 200 GeV?

Radial response (n = 0)

Does $\kappa 0(p_T)/\kappa 0$ stay universal at low p_T and separate at higher p_T?

Conclusions: is opacity enough?

- Equal η/s does not fix finite-opacity anisotropic flow
- At leading opacity, normalized harmonic ratios measure relaxation-rate ratios, independent of geometry
- For the QCD kernel, each angular sector has a momentum-dependent spectrum, given in closed form
- Dilute flow, viscosity and late-time decay probe three projections of one operator
- Preliminary SMASH data show opposite opacity trends in R_{32} for two models



No: the collision spectrum matters, and it has to be resolved to be tested.

Thank you!

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CRC-TR 211 Strong-interaction matter under extreme conditions

HFHF

Back up

How is the conservation built into the collisional operator?

- $\boldsymbol{\gamma_0 = \gamma_1 = \gamma_{-1} = 0}$**:** the 0th and 1st hormonics are the conserved energy and momentum densities
- The naive spectrum $\mathbf{D}\,\boldsymbol{\ell^2}$ is not momentum conserving, kept as reference
- Corrected spectrm , with the conserved sectors projected out
- Fokker-Planck kernel: conserved sectors frozen(see the comparison for full projection in supplemental material of 2609.05703)

Back up

Why high angular modes matters?

- Flat RTA gives $D32 = 1$: no angular structure of κ_3/κ_2
- Spectra with angular structure gives $D32 \sim 1.5\text{-}2.2$, so $\ell = 3$ rate changes the triangular response
- η/s constrains only γ_2, while the higher rates set the response at finite opacity
- Late time relaxation is governed by the slowest mode with nonzero overlap with the source

	2D		3D	
	l^2	l^2-1	$l(l+1)$	$l(l+1)-2$
γ_3/γ_2	9/4	8/3	2	5/2
γ_4/γ_2	4	5	10/3	9/2

Relevance for triangular flow.
A single opacity flat RTA can not reproduce the κ_3/κ_2 spectral dependence -> the angular structure of rates is needed